

Individual Assignment 5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

Question 1

A: Planning

- x axis: Site

- y axis: log transformed Ostracod abundance
- geometry abundance: geom_jitter()
- geometry median: stat_summary()

B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
2	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA
3	2023-05-04_NEC	2023-05-04 00:00:00	NEC	ostr~	0.667	8.05	NA	8.55
4	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.7	NA	NA	12.1
5	2023-09-15_NMC	2023-09-15 00:00:00	NMC	ostr~	0.4	NA	NA	14.2
6	2021-04-15_NPB	2021-04-15 00:00:00	NPB	ostr~	185.	8.09	3.57	8.24
7	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	ostr~	0	NA	NA	NA
8	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
9	2023-11-12_NMC	2023-11-12 00:00:00	NMC	ostr~	0.533	NA	47	14.0
10	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
 # Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
 # Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
 # comments <chr>, site_full <fct>, log_ave_lit <dbl>

C: Code Figure

```
# base layer
ostracods_plot <- ggplot(data = ostracod,
```

```

        # x-axis: site
        # y-axis: mean abundance/L
        mapping = aes(x = site_full,
                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracods_plot

```

